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### IMAGE LABEL SHARING SYSTEM

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#### Abstract

Various systems and methods are presented regarding segmentation of medical images whereby a segmentation process comprises of channels configured to share labels. Accordingly, rather than each label in a series of images all requiring an individual channel in a segmentation model, the segmentation model can be configured such that a single channel (e.g., having a single group of labels) is shared by images having multiple labels. By sharing a channel, and label group, across multiple labels, the efficiency of the segmentation model is improved as fewer channels are required to segment a series of images. Matching between regions of interest across the series of images can be performed to determine a number of shared channels required for the segmentation model. The series of images can be further applied to the segmentation model to generate a set of labeled segmented images.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to India Provisional Patent Application No. 202441015934 filed on Mar. 6, 2024, entitled “IMAGE LABEL SHARING SYSTEM”. The entireties of the aforementioned application are incorporated by reference herein.

### TECHNICAL FIELD

[0002] This application relates to systems and techniques facilitating segmentation imaging in a biomedical imaging system.

### BACKGROUND

[0003] Segmentation tasks and deep learning methods can be utilized for biomedical imaging applications, whereby, such tasks and methods can involve analyzing/processing images of multiple anatomies. As part of such processes, each anatomy image can be annotated with one or more labels identifying one or more regions of interest. Labelling of the regions can be a complex task requiring multiple parameters, and/or channels, in accordance with the number of images and associated labelled regions.

[0004] The above-described background is merely intended to provide a contextual overview of some current issues and is not intended to be exhaustive. Other contextual information may become further apparent upon review of the following detailed description.

### SUMMARY

[0005] The following presents a simplified summary of the disclosed subject matter to provide a basic understanding of one or more of the various embodiments described herein. This summary is not an extensive overview of the various embodiments. It is intended neither to identify key or critical elements of the various embodiments nor to delineate the scope of the various embodiments. The sole purpose of the Summary is to present some concepts of the disclosure in a streamlined form as a prelude to the more detailed description that is presented later.

[0006] In one or more embodiments described herein, systems, devices, computer-implemented methods, configurations, apparatus, and/or computer program products are presented to generate segmented images utilizing a segmentation model having fewer output channels than utilized with a conventional approach.

[0007] According to one or more embodiments, a system is presented, wherein the system comprises at least one processor, and at least one memory coupled to the at least one processor and having instructions stored thereon, wherein the system can be configured to generate segmented images from a segmentation model utilizing a number of channels matching a number of common label groups. In response to the at least one processor executing the instructions, the instructions facilitate performance of operations, comprising receiving a set of images, wherein the set of images comprises: a first image having a first region of interest (RoI), wherein the first image is associated with a first task, and a second image having a second RoI, wherein the second image is associated with a second task, and the first RoI and the second RoI are included in a common shared label group. In a further embodiment, the operations can further comprise applying the set of images to a segmentation model, wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel configured to output segmented images based the common shared label group. In another embodiment, the operations can further comprise outputting, based on the common shared label group, at least one segmented image comprising the first RoI or the second RoI.

[0008] In an embodiment, the common label group is a first label group, the output channel is a

first output channel, and at least one segmented image is included in a first set of segmented images, wherein the operations further comprise receiving a second set of images sharing a second shared label group, wherein the first shared label group and the second shared label group are disparate, and the second set of images comprises: a third image having a third RoI, wherein the third image is associated with the first task and a fourth image having a fourth RoI, wherein the fourth image is associated with the second task, and the third RoI and the fourth RoI are included in a second shared label group. Further applying the second set of images to the segmentation model, wherein the segmentation model further comprises a second output channel configured to output segmented images based on the second shared label group, and further outputting, based on the second shared label group, at least one segmented image comprising the third RoI or the fourth RoI.

[0009] In another embodiment, an architecture of the segmentation model is common to application of two or more segmentation tasks and generation of segmented images based on the first shared label group and the second shared label group.

[0010] In a further embodiment, a set of common weightings are applied during creation of both the first shared label group and the second shared label group during application of two or more segmentation tasks.

[0011] In another embodiment, labels in the first shared label group are disparate to the labels in the second shared label group.

[0012] In another embodiment, the operations can further comprise determining a first number of shared label groups in the first task and a second number of shared label groups in the second task, and further, in response to a determination that the first number of shared label groups is different to the second number of shared label groups, configuring the number of output channels of the segmentation model to equal the greater of the first number of shared label groups or the second number of shared label groups.

[0013] In an embodiment, the operations can further comprise generating the common shared label group as a function of at least one similarity between the at least two images.

[0014] In another embodiment, the operations further comprise applying a third task to the segmentation model, further generating at least one shared label group from the third task, wherein the at least one shared label group includes the common shared label group, and further training the segmentation model across multiple tasks incrementally utilizing any of the at least two images or the common shared label group.

[0015] In a further embodiment, the operations can further comprise subsequently applying a subsequent task to the segmentation model to facilitate incremental training of the segmentation model over a series of iterations to adapt the segmentation model to the subsequent task, and maintaining incremental training until determining convergence has occurred.

[0016] In another embodiment, the operations can further comprise re-applying a prior task to the converged segmentation model to prevent catastrophic forgetting of the prior task as a function of the segmentation model being converged with the subsequent task.

[0017] In further embodiments, a computer-implemented method is provided, wherein the method comprises receiving, by a device comprising at least one processor, a first image comprising a first segmentation label associated with a first region of interest (RoI), wherein the first segmentation label is included in a first shared label group. The operations can further comprise receiving, by the device, a second image, wherein the second image comprises a second RoI and a second segmentation label associated with the second RoI, further determining, by the device, whether the second RoI relates to the first RoI, and further, in response to determining, by the device, the second RoI is related to the first RoI, applying, by the device, the second segmentation label to the first shared label group.

[0018] In an embodiment, the first shared label group is associated with a first channel assigned to a segmentation model implemented by the device, wherein the segmentation model is implemented

by the device to determine the second RoI is related to the first RoI.

[0019] In another embodiment, in response to determining by the device, the second RoI is unrelated to the first RoI, the computer-implemented method further comprising identifying, by the device, a third image comprising a third segmentation label, wherein the third segmentation label is associated with a third RoI and the third segmentation label is included in a second shared label group. Further determining, by the device, the second RoI relates to the third RoI, and further applying the second segmentation label to the second shared label group, wherein the second shared label group is associated with a second channel assigned to the segmentation model implemented by the device, wherein the segmentation model is implemented by the device to determine the second RoI is related to the third RoI.

[0020] In another embodiment, in response to determining by the device, the second RoI is unrelated to the first RoI and is also unrelated to the third RoI, the computer-implemented method further comprising updating, by the device, the segmentation model to include a third channel, wherein the third channel is assigned with a third shared label group, and further assigning the second segmentation label to the third shared label group.

[0021] In an embodiment, the segmentation model can be a U-net model.

[0022] In another embodiment, wherein the determination, by the device, of the second RoI relates to the first RoI is based on at least one of similarity of shape of the first RoI to the second RoI, pixel level between the first RoI and the second RoI, or patch level between a first region of interest (RoI) and a secondRoI.

[0023] Further embodiments can include a computer program product stored on a non-transitory computer-readable medium and comprising machine-executable instructions, wherein, in response to being executed, the machine-executable instructions cause a system to perform operations, comprising receiving a set of images, wherein the set of images comprises a first image having a first region of interest (RoI), wherein the first image is associated with a first task, and a second image having a second RoI, wherein the second image is associated with a second task, and the first RoI and the second RoI are included in a common label group. Further applying the set of images to a segmentation model, wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel configured to output segmented images based the common label group, and further outputting, based on the common shared label group, at least one segmented image comprising the first RoI or the second RoI.

[0024] In an embodiment, the set of images in the task comprise medical images.

[0025] In another embodiment, the first RoI being related to the second RoI is based on at least one of similarity of shape of the first RoI to the second RoI, pixel level between the first RoI and the second RoI, or patch level between a first RoI and a second RoI.

[0026] In a further embodiment, the operations can further comprise defining the number of output channels prior to generating the set of shared label groups.

## BACKGROUND

[0027] Technological advancements in the last years have prompted the medical world to strive for precision healthcare. The goal of precision healthcare comprises the ability for a caregiver to retrieve precise information (about the patient and/or about treatment) at the right time and place. One way in which this ability is needed relates to the time available to review a patient's medical data tends to become shorter as the workload and working requirements of medical staff increases (e.g., as radiologists and clinicians work intensifies). Another way is the need for ready and easy querying sources of information, with accurate/well considered treatment recommendations.

[0028] The above-described background is merely intended to provide a contextual overview of some current issues and is not intended to be exhaustive. Other contextual information may become further apparent upon review of the following detailed description.

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## Description

### DESCRIPTION OF THE DRAWINGS

[0029] One or more embodiments are described below in the Detailed Description section with reference to the following drawings.

[0030] FIG. 1 presents a high-level overview of a segmentation system utilizing a shared-label configuration, in accordance with one or more embodiments.

[0031] FIG. 2 presents a schematic illustrating a segmentation model, in accordance with one or more embodiments.

[0032] FIG. 3 presents a sample shared label space comprising a series of shared labels for segmentation of respective example anatomies, in accordance with an embodiment.

[0033] FIG. 4 illustrates a high-level view of operation of a shared label segmentation system, in accordance with one or more embodiments.

[0034] FIG. 5 presents a schematic illustrating an example operation of a label match component, in accordance with an embodiment.

[0035] FIG. 6 presents a sequence of tasks generated by an incrementally trained segmentation model, in accordance with an embodiment.

[0036] FIG. 7 presents a set of output images from combined model inference output, after incremental learning has been performed.

[0037] FIG. 8 illustrates a computer-implemented method for applying a prompt to a question to constrain scope of question and/or supplied answer, in accordance with an embodiment.

[0038] FIG. 9 illustrates an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0039] FIG. 10 illustrates an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0040] FIG. 11 illustrates an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0041] FIG. 12 is a block diagram illustrating an example computing environment in which the various embodiments described herein can be implemented.

[0042] FIG. 13 is a block diagram illustrating an example computing environment with which the disclosed subject matter can interact, in accordance with an embodiment.

### DETAILED DESCRIPTION

[0043] The following detailed description is merely illustrative and is not intended to limit embodiments and/or application or uses of embodiments. Furthermore, there is no intention to be bound by any expressed and/or implied information presented in any of the preceding Background section and/or in the Detailed Description section.

[0044] One or more embodiments are now described with reference to the drawings, wherein like referenced numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a more thorough understanding of the one or more embodiments. It is evident, however, in various cases, that the one or more embodiments can be practiced without these specific details.

[0045] It is to be understood that when an element is referred to as being “coupled” to another element, it can describe one or more different types of coupling including, but not limited to, chemical coupling, communicative coupling, electrical coupling, electromagnetic coupling, operative coupling, optical coupling, physical coupling, thermal coupling, and/or another type of coupling. Likewise, it is to be understood that when an element is referred to as being “connected” to another element, it can describe one or more different types of connecting including, but not limited to, electrical connecting, electromagnetic connecting, operative connecting, optical connecting, physical connecting, thermal connecting, and/or another type of connecting.

[0046] As used herein, “data” can comprise metadata. Further, ranges A-n and A-i are utilized herein to indicate a respective plurality of devices, components, signals etc., where n and i are any positive integers.

[0047] The various embodiments presented herein relate to a technique/method/process called “label-sharing”. As further described, each label (aka a channel) included each respective task is shared across independent segmentation tasks, such that a first label in a first task shares a channel with a second label in a second task. In an aspect, segmentation and segmentation labels relate to identifying/labeling pixels of interest versus pixels not of interest in an image.

[0048] As previously mentioned, segmentation tasks and deep learning methods can be utilized for biomedical imaging applications, whereby, such tasks and methods can involve analyzing/processing images of multiple anatomies. As part of such processes, each anatomy image can be annotated with one or more labels identifying one or more regions of interest. Labelling of the regions can be a complex task requiring multiple parameters, and/or channels, in accordance with the number of images and associated labelled regions.

[0049] Present approaches for multi-task incremental learning include (a) multi-task learning; (b) domain adaptation approaches; (c) mixture of experts; and (d) Foundation models. Approaches (a) and (c) requires a large number of output channels and, in most cases, approaches (a) and (c) do not necessarily share labels. Domain adaptation on the other hand has serious issues of catastrophic forgetting and suboptimal solution across tasks. Foundation models, while powerful, are excessively heavy, resulting in high inference costs and may not be optimal for specific task sets, as they are designed for more generalized application.

[0050] In the area of incremental learning, numerous solutions leverage textual descriptions delineating objects or regions of interest, alongside their associated embeddings, as input to a vision model for the detection of novel items or the execution of new tasks.

[0051] However, these methods rely on the availability of pretrained foundation models during inference which may not be feasible or effective for medical imaging.

[0052] In one approach, separate models can be trained for each anatomy. In another approach, a common block model for learning general features (e.g., by an encoder) followed by separate task heads for each anatomy/task (e.g. by a decoder) can be utilized. However, both approaches require additional parameters for each new task.

[0053] Another conventional approach for mitigating the requirement of additional parameters for each new task, is to use a single, common model but with different labels (also known as channels) across different anatomies. However, adoption of this approach requires an increase in the total number of channels (and parameters) for each new anatomy to be detected by the common model.

[0054] Further, with conventional approaches, in the event of a model already being trained and further performing an incremental learning process, implementation of an additional task(s) can be difficult/impossible, due to one or more changes required in the model architecture to implement the additional task(s).

#### Review of a Conventional Approach

[0055] Regarding a conventional approach, in conjunction with two example operational scenarios (i) and (ii): [0056] a) let there be ‘N’ independent tasks, where independent indicates that there is no overlap in the set of images used for any of the tasks, e.g., imagery of a hand cannot appear in both a first task and a second task concurrently applied to a segmentation model. [0057] b) let the tasks be  $T_{sub.i}$ ,  $i=1, \dots, N$ . [0058] c) let the number of segmentation labels for  $T_{sub.i}$  be  $n_{sub.i}$ ,  $L_{sub.ij}$  being the  $j$ .sup.th landmark/segmentation mask for task  $T_{sub.i}$ .

[0059] Accordingly, per Equation 1, the total number of labels S to be learnt by a segmentation model or a set of models will be

$$[00001] S = \sum_{i=1}^N n_i \quad \text{Eqn. 1}$$

[0060] Generally, the two operational scenarios are: (i) a network has S number of channels such

that, based on the choice of tasks, the set of channels are chosen as an output, or (i) a network is retrained to adapt to a new task. With the conventional approach, scenario (i) becomes unmanageable after a limited set of tasks and beyond a limited number of labels since S tends to become prohibitively large, and further, with scenario (ii), catastrophic forgetting becomes challenging.

### Label Sharing System

[0061] Per the various embodiments presented herein, a single, label sharing model is utilized with all of the parameters remaining the same for all of the respective tasks. Further, the label sharing model can be trained with the labels across multiple tasks being shared per respective label groups. For example, label L.sub.13 identifying a first region of interest (RoI) in a first image can be in the same shared label group as a label L.sub.21 identifying a second RoI in a second image, and so on, as further described. Hence, with the various embodiments presented herein, contrary to the conventional approach (per Equation 1) the label sharing model schema only requires, per Equation 2:

[00002]  $G = \max_{i=1, \text{Math. } N} (n_i) \text{ number of labels or channels}$  Eqn. 2 [0062] where G means grouping, as further described.

### Benefits of Utilizing a Label Sharing System

[0063] Utilizing a label sharing model as part of a segmentation process has multiple benefits over conventional approaches, such as, in a non-limiting list:

[0064] a) reduction in the number of channels from  $\Sigma_{\text{sub.1.sup.Nn.sub.i}}$  to

[00003]  $\max_{i=1, \text{Math. } N} (n_i)$ , per Equations 1 and 2, [0065] b) maintains a consistent set of channels

across tasks, [0066] c) is lightweight, [0067] d) optimized for specific tasks without the potential for catastrophic forgetting, [0068] e) enables incremental learning for additional tasks without having to change the model architecture, [0069] f) improved speed of execution and extends existing models for implementation with additional tasks, [0070] g) can be utilized across multiple/disparate products and use-cases of multi-label segmentation tasks, [0071] h) modality and domain agnostic, [0072] i) enables a training methodology to be implemented for incremental learning that ensures no forgetting of past tasks, and [0073] j) automatically determines the labels to be shared based on landmark similarity and utilizing a matching optimization procedure.

[0074] Accordingly, the various embodiments presented herein disclose and facilitate, per (A)-(D):

[0075] A) A novel framework of a unified model for label-sharing across different multi-label segmentation tasks. The various embodiments utilize a single model architecture configured to share output labels across multiple tasks, enabling incremental learning without altering the number of output channels. Consequently, each output channel within the network is utilized for sharing labels across various tasks. [0076] B) Transitioning from one task to another may involve training the network with the new dataset for a few epochs while selecting the suitable channels for label sharing. For instance, if the network has already been trained for M tasks, incorporating the (M+1).sup.th task entails initially training the pre-trained network for the new task, followed by additional epochs of retraining that include the previous tasks as well. Such an approach mitigates the risk of catastrophic forgetting, a prevalent issue (e.g., with a conventional approach) when introducing a new task. [0077] C) In the context of a single network addressing multi-label segmentation across multiple tasks, the total number of output channels for the network is determined by

[00004]  $\max_{i=1, \text{Math. } N} n_i$ , [0078] where i represents the task number and n.sub.i is the labels

corresponding to i.sup.th task. This differs significantly from the conventional requirement of  $\Sigma_{\text{sub.i=1.sup.Nn.sub.i}}$ , to stack different multi-label segmentation tasks. [0079] D) Design of the model architecture(s) presented herein take into consideration the selection of channels for label sharing. Various systems and methods are presented for determining the optimal label-channel

combinations across different tasks, aiming to share the most suitable set of labels for a given channel. The various systems and methods can utilize similarity or distance metrics calculated by (a) Foundation Model (FM) features, (b) hand-crafted features, (c) other network-based features (e.g., deep learning-network-based features), (d) employ techniques such as simulated annealing or similar methods, and suchlike. The computation of similarity can be tailored to a specific use case. The label-sharing mechanism remains resilient even when faced with a diverse range of labels.

[0080] It is to be appreciated that while the various embodiments presented herein are directed to biomedical applications, the embodiments are not so limited and can be equally applied to any application utilizing image segmentation or other applicable technology/techniques.

[0081] Accordingly, per the various embodiments presented herein, a number of common labels can be determined for a set of images in a task, wherein the various images in the set of images can include regions of interest (RoIs). The number of common labels can be established based on the number of RoIs, assigned/identified labels, etc. The number of common labels can then be applied to a segmentation model, whereby the number of common labels defines the number of output channels of the segmentation model. Further images can be subsequently processed and further utilized to confirm that the segmentation model is performing correctly with the current combination of common labels and output channels. In the event of the segmentation model is not performing as expected, the number of common labels can be reassessed and the segmentation model updated accordingly.

[0082] FIG. 1, system 100, presents a high-level overview of a segmentation system utilizing a shared-label configuration, in accordance with one or more embodiments. As shown in FIG. 1, a segmentation system 110 can be configured to generate parameter optimized shared labels for independent multi-anatomy segmentation and incremental learning. Segmentation system 110 can be configured to receive a set of tasks 150A-n, wherein, per the various embodiments presented herein, tasks 150A-n can comprise a set of images 160A-n (a.k.a., task images) wherein each image in the set of images 160A-n has been previously annotated with one or more labels 165A-n indicating respective regions of interest (RoI) 162A-n on the respective images 160A-n. Labels 165A-n for RoIs 162A-n can be generated by subject matter experts, e.g., by a human entity annotating the images 160A-n (e.g., using a digital pen), and can further provide ground truth information to a supervised machine learning algorithm (e.g., segmentation model 125A-n, as further described).

[0083] Segmentation system 110 can comprise any suitable architecture and further include a segmentation component 120 configured to perform biomedical image segmentation. In an embodiment, segmentation component 120 can be configured to utilize one or more segmentation models 125A-n. Any suitable segmentation model 125A-n can be utilized for image segmentation of the various tasks 150A-n. For example, segmentation model 125A-n can be an artificial neural network, a convolutional neural network, a fully convolutional network, a diffusion model, and suchlike. Tasks 150A-n comprising the images 160A-n, RoIs 162A-n, and labels 165A-n can be utilized by the segmentation component 120 to train the segmentation model 125A-n.

Segmentation model 125A-n can be provided by process component 170, wherein segmentation model 125A-n can be included in processes 176A-n, as further described.

[0084] In an example embodiment, segmentation model 125A-n can be a convolutional neural network having a U-net configuration, e.g., a U-shaped encoder/decoder architecture. Turning briefly to FIG. 2, schematic 200 illustrates segmentation model 125A-n having an example U-net architecture, wherein the term “architecture” relates to an arrangement of deep learning tools such as convolutional layer, max pooling, etc. As shown, the example U-net architecture can be configured with a contraction/encoder path comprising a series of encoder blocks 210A-n (e.g., convolutional layers and intermediate max pool layers) and an expansion/decoder path comprising a series of decoder blocks 220A-n (e.g., convolutional layers and intermediate up sample operations), where, in the presented example configuration, U-net architecture 125A comprises four



encoder blocks **210A-D** and four decoder blocks **220D**.

[0085] In an example operation, the U-net architecture of segmentation model **125A-n** can receive a set of input tasks **150A-n**, for example, a first task **150A** (e.g., image **160A** of a foot, axial view), a second task **150B** (e.g., image **160B** of a knee), a third task **150C** (e.g., image **160C** of a hip), a fourth task **150D** (image **160D** of a foot, sagittal view), and suchlike. Utilizing the encoder blocks **210A-n**, U-net architecture **125A** can be configured to extract one or more useful features, RoIs **162A-n**, from the input image **160A-n**, utilizing multiple convolutional layers. Decoder blocks **220A-n** are configured to up sample the one or more RoIs **162A-n** (e.g., using a transpose convolution operation in conjunction with a concatenation process). As further shown, the U-net architecture **125A** generates/outputs respective segmentation masks **240A-n**, e.g., demarking/outlining RoIs **162A-n**. In an aspect, U-net architecture **125A** can utilize connecting paths **250A-n** to concatenate an output of an encoder layer as the input of a decoder layer, feeding features from one part of the encoder blocks **210A-n** to decoder blocks **220A-n**.

[0086] In an embodiment, as previously mentioned, during training, tasks **150A-n** and respective RoIs **162A-n** on the sample images **160A-n**, can be labelled/annotated with labels **165A-n**. When using a conventional approach, each label **165A-n** on each sample image **160A-n** requires an independent channel **260A-n** as an output of the segmentation model **125A-n**. Per the various embodiments presented herein, by sharing labels **165A-n** across the RoIs **162A-n**, images **160A-n** and tasks **150A-n**, the number of channels **260A-n** required for outputs of the segmentation model **125A-n** can be reduced. And further, when utilizing a segmentation model **125A-n** with parameter optimized shared labels **165A-n**, incremental learning can be implemented. With the segmentation model **125A-n** trained with tasks **150A-n**, sample images/tasks **195A-n** can be applied to the segmentation model **125A-n**, with trained segmentation model **125A-n** further configured to generate/output segmented images **240A-n**. It is to be appreciated that given the limits of rendering images in black and white, the respective shading of the RoIs **162A-n** in the various FIGS. 2-4, 6, and 7 is not consistent and arbitrary shading is being utilized to convey distinct RoIs **162A-n**, labels **165A-n**, being implemented on respective images in accordance with the various embodiments presented herein.

[0087] FIG. 1 presents a series of steps **1-4B** presenting a high-level overview of a sequence of operations that can be implemented with segmentation system **110** and system **100** generally, the steps are further described per FIG. 4.

[0088] At step **1**, a series of tasks **150A-n** can be received at segmentation system **110**, wherein the tasks **150A-n** include a collection of sample images **160A-n** having RoIs **165A-n**, wherein the respective RoIs **165A-n** are respectively labelled with labels **165A-n**.

[0089] At step **2**, the tasks **150A-n** are applied to the match component **130** configured to match respective labels **165A-n** into a collection of label groups **157A-n** (e.g., per FIGS. 3 and 5).

[0090] At step **3A**, the tasks **150A-n** with labels **165A-n** in images **160A-n** respectively assigned to the two or more label groups **157A-n** are applied to the segmentation model **125A**, wherein segmentation model **125A** generates/outputs the segmented images **196A-n** per the respective label groups **157A-n**.

[0091] At step **3B**, the segmented images **196A-n** can be reviewed for correctness (e.g., correctly labelled RoIs **162A-n** and label groupings **157A-n**). In response to determining the segmented images **196A-n** are as expected, the match component **130** and segmentation model **125A** can be considered to be operating correctly/trained, e.g., the determined number of group labels **157A-n** generated by the match component **130** enables the segmentation model **125A** to correctly generate/label the segmented images **196A-n**. In the event of the segmented images **196A-n** are not as expected, steps **1-3B** can be re-performed.

[0092] At step **4A**, sample tasks/images **195A-n** (e.g., comprising actual data/RoIs/labels comparable to RoIs **162A-n** and labels **165A-n**) can be applied to the trained segmentation model **125A**, wherein the segmentation component **125A** generates segmented images **196A-n** from the

sample images **195A-n**.

[0093] At step **4B**, the segmented images **196A-n** generated from the sample images **195A-n** can be reviewed for correctness. In response to a determination that the segmented images **196A-n** are being generated correctly, steps **4A** and **4B** can be repeated. In response to a determination that the segmented images **196A-n** are being incorrectly generated (e.g., the number of label groups **157A-n** is no longer sufficient) the match component **130** and/or the segmentation model **125A-n** can be retrained.

[0094] As further shown in FIG. **1**, segmentation system **110** can be communicatively coupled to/include a computer system **180**. Computer system **180** can include a memory **184** that stores the respective computer executable components (e.g., segmentation component **120**, match component **130**, process component **170**, and suchlike, as further described herein) and further, a processor **182** configured to execute the computer executable components stored in the memory **184**. Memory **184** can be further configured to store any of respective tasks **150A-n**, images **160A-n**, RoIs **162A-n**, labels **165A-n**, processes **176A-n**, sample images **195A-n**, segmented images **196A-n**, label groups **157A-n**, models **125A-n**, matrix **410**, vector pairings **510A-n**, and suchlike.

[0095] The computer system **180** can further include a human machine interface (HMI) **186** (e.g., a display, a graphical-user interface (GUI)) which can be configured to present various information including tasks **150A-n**, images **160A-n**, labels **165A-n**, model **125A-n**, label groups **157A-n**, vector pairings **510A-n**, and suchlike, per the various embodiments presented herein. HMI **186** can include an interactive display/screen **187** to present the various information. Computer system **180** can further include an I/O component **188** to receive and/or transmit respectively newly defined tasks **150A-n**, images **160A-n**, RoIs **162A-n**, labels **165A-n**, model **125A-n**, label groups **157A-n**, vector pairings **510A-n**, and suchlike. Any suitable technology can be utilized for interaction/communication by I/O **188**, e.g., file transfer protocol (FTP), simple radio standalone (SRS), and suchlike.

#### Examples of Shared Labelling Across Series of Anatomy Images

[0096] FIG. **3** presents an example sample shared label space **300** comprising a series of shared labels **157A-n** for segmentation of respective example anatomies. FIG. **3** presents a set of example tasks **150A-n** comprising images **160A-n** with labels **165A-n** (RoIs **162A-n** identified). The example tasks **150A-n** are presented horizontally, such that images **160A-n** pertaining to a first task are presented in FIG. **3** on the horizontal. For example, images **160A** to **160R** on the horizontal pertain to foot axial task **150A**. Further, images **160** to **160n** on the horizontal pertain to knee task **150B**, hip task **150C** comprises images **160C** to **160o**, and foot sagittal task **150n** comprises images **160D** to **160P**.

[0097] Further shown in FIG. **3** are a series of shared label groups **157A-n**, aligned vertically in FIG. **3**. For example, eight shared label groups **157A-n** are utilized across the set of tasks **150A-n**. For example, a first shared label **157A** (red) is used to label/annotate a respective region on each of tasks **160A-n**. Shared Label **157B** (green) is used to label regions on both foot axial and hip tasks **150A** and **150n**. Shared Label **157C** (blue) is used to label regions on foot axial, knee, and hip tasks **150A**, **150B**, and **150C**. Shared Label **157D** (purple) is used to label regions on foot axial, knee, hip, and foot sagittal tasks **150A-n**. Shared Label **157E** (orange) is used to label regions on foot axial, hip, and foot sagittal tasks **150A**, **150C**, and **150n**. Shared Label **157F** (light green) is used to label a region on an image in the foot axial task **150A**. Shared Label **157G** (mauve) is used to label a region on an in foot axial task **150A**. Shared Label **157H** (peach) is used to label regions on foot axial and knee tasks **150A** and **150B**.

[0098] The concept of common labelling is further presented per the fifth column having the shared label **157E**, whereby tasks **150A**, **150C**, and **150n** are presented with corresponding RoIs **162A-C** and labels **165A-C** indicated. While not shown for the other columns of shared labels **157A-D** and **157F-n**, similar labeling **165A-n** for tasks **150A-n** can be applied to the respective RoIs **162A-n**.

[0099] Accordingly, as presented in FIG. **3** and as further described, only eight shared label groups

**157A-H** and a corresponding eight channels **260A-n** (in the segmentation model **125A**) are required. However, with a conventional approach, where each distinct region **162A-n** on each distinct image **160A-n** would require a unique label and an associated channel, at least 20 channels would be required as outputs of the segmentation model **125A-n**, where each channel corresponds to each label in the twenty tasks **150A-n**.

#### Example Process Workflow

[0100] FIG. 4, schematic **400** illustrates a high-level view of operation of a shared label segmentation system, in accordance with one or more embodiments.

[0101] A series of tasks **150A-n** (and associated images **160A-n**) are provided to a match component **130**, wherein match component **130** can be configured to group individual labels **165A-n** of respective images **160A-n** in tasks **150A-n** into the label groups **157A-n** (also referred to as LG **1-n**). As shown, match component **130** can utilize a database/matrix **410**, wherein the matrix **410** can be populated by the match component **130**. In the example presented: [0102] Label Group **1/157A** comprises: label **L4/165D** of task **1** (task **150A**), label **L1/165Z** of task **2** (task **150B**), label **L5/165P** of task **3** (task **150C**), and label **2/165S** of task **4** (task **150D**); [0103] Label Group **2/157B** comprises: label **L3/165C** of task **1**, label **L3/165R** of task **2**, label **L2/165F** of task **3**, and label **L1/165N** of task **4**; and [0104] Label Group **3/157C** comprises: label **L2/165B** of task **1**, label **L2/165T** of task **2**, and label **L4/165H** of task **3**.

[0105] In an example embodiment, the respective steps in the process workflow are: [0106] a) let the tasks **150A-n** be T.sub.i, i=1, . . . , N and let the number of segmentation labels for each T.sub.i be n.sub.i, with L<sub>y</sub> being label corresponding to the j.sup.th landmark for i.sup.th task. [0107] b) a sample data from the training set consisting of M representative samples (e.g., images **160A-n**) along with their different segmentation labels (e.g., labels **165A-n**) is presented to the match component **130**. Match component **130** is configured to determine an optimal matching between respective labels **160A-n** for different tasks **150A-n**. In an embodiment, match component **130** can be configured to operate in conjunction with process component **170** utilizing one or more processes **176A-n** to perform the similarity matching (e.g., between and/or across RoIs **162A-n**). Different options for design and operation of the match component **130** are further described in FIG. 5. [0108] c) In an embodiment, match component **130** can be further configured to generate output of a set of label groups **157A-n**, where a label group denotes a respective RoI **162A-n** for each task **150A-n** having the same shared label **157A-n**. Parameter (denotes a label group **157A-n**, wherein the number of label groups is

[00005]  $G = \max_{i=1 \dots N} (n_i)$ . [0109] d) in a further embodiment, based on assigning the same shared labels **157A-n** for each task **150A-n**, and applying the one or more tasks **150A-n** to a segmentation component **120**, segmentation model **125A-n** can be generated/trained. In a further embodiment, the segmentation component **120** and segmentation model **125A-n**, configured with a number G output channels **260A-n**, can be trained. [0110] e) during an inference time period, the interpretation of the shared labels **157A-n** by the segmentation component **120** is performed with the assumption that the input task type is known.

#### Example Operation of a Match Component

[0111] FIG. 5, schematic **500**, illustrates an example operation of match component **130**, in accordance with an embodiment. As previously mentioned, match component **130** can be configured to match labels **165A-n** (pertaining to respective images **160A-n** in tasks **150A-n**) and further group the matched labels **165A-n** in label groups **157A-n**. Match component **130** can be configured to receive a set of tasks **150A-n** comprising respective images **160A-n** respectively labelled with labels **165A-n** pertaining to RoIs **162A-n**. In an embodiment, match component **130** can be further configured to perform similarity analysis (or any other applicable technology) on the respective labels **165A-n** and associated RoIs **162A-n** across the respective images **160A-n** in tasks **150A-n**, to generate respective similarity measures **510A-n** (e.g., measures 0.63, 0.74, 0.91, 0.45,

etc., between respective labels **165A-n**), and based thereon, can further group matched labels **165A-n** to generate label groups **157A-n**. While not shown, match component **130** can be further configured to apply weightings during generation of the similarity measures **510A-n**, e.g., to capture relationships between labels **165A-n** during creation of the respective label groups **157A-n**. For example, labels **165A-n** RoIs **162A-n** may have a common feature such as sample images **160A-n** were generated on a common system, were not generated on a common system, RoIs **162A-n** and labels **165A-n** were generated by a common entity (e.g., a common system, the same human reviewer), and suchlike. Per the following schemes 1-3, when, for example, a common U-net based architecture is trained for all tasks **150A-n** but each label **165A-n** from each task **150A-n** retains the assigned unique label **165A-n** (e.g., per a conventional, non-label sharing, unique labeling approach), the number of output channels **260A-n** would therefore be  $\sum_{i=1}^{sub.N} sub.i$ .

[0112] The following now describes three schemes, Schemes 1-3 utilizing the parameter optimized shared label approach, per the various embodiments presented herein.

Scheme 1

[0113] M exemplar images **160A-n** containing each of the RoIs **162A-n**/labels **165A-n** for each of the tasks **150A-n** is determined and the feature vectors (not shown) of each of these exemplar images **160A-n** is derived as the intermediate representation of an encoder on one of the decoder layers.

[0114] For a label pair 'a' for T.sub.i and 'b' for T; where  $i \neq j$ , the respective average similarity **510A-n** of these feature vectors between images **160A-n** belonging to each label group **157A-n** is determined (e.g., by match component **130**).

[00006]  $Match(T_{ia}, T_{jb}) = \frac{\text{mean}_{m \in \{1, \dots, M\}} \text{sim}(\text{feat}(T_{ia}^m), \text{feat}(T_{jb}^n))}{n \in \{1, \dots, M\}}$

where sim is the cosine similarity and feat is the features extracted from the encoder or decoder of the initial trained network.

[0115] The similarity **510A-n** for different labels **165A-n** across tasks **150A-n** can also be derived in the following manner:

Scheme 2:

[0116] Match component **130** can utilize an object detection model (e.g., process **176A-n**, a Foundation Model (FM)) technique, configured for feature extraction at both patch and pixel levels. Accordingly, an object detection model **176A** can be configured to recognize RoIs **162A-n** in images **160A-n**. When segmentation masks (e.g., comparable to segmentation masks **240A-n**) are provided, the features of pixels (e.g., of respective RoIs **162A-n**) associated with each segmentation mask within the image **160A-n** can be aggregated to generate a feature vector representation (not shown) for a specific label **165A-n**. Subsequently, a similarity-based method, as previously described, can be employed to ascertain matches between various RoIs **162A-n** and labels **165A-n** across diverse tasks **150A-n**.

Scheme 3:

[0117] Utilizing hand-crafted features like shape, texture or other image-based features corresponding to each of the RoIs **162A-n**/labels **165A-n** across the tasks **150A-n** can be used by match component **130** to find similarity **510A-n** between labels **165A-n** and use similar formulation to that presented in Scheme 1.

Label Matching Based on Similarity of RoI of Different Tasks

[0118] Once the respective similarity match degrees **510A-n** have been determined between every pair of labels **165A-n** across different tasks **150A-n**, an optimal matching of labels **165A-n** across different tasks **150A-n** can be performed by match component **130**. Any suitable technology/process **176A-n** can be utilized by match component **130**, e.g., a Hungarian algorithm for two tasks or for more complex scenarios, variants such as Integer Programming, Genetic Algorithms, Simulated Annealing for multiple sets, and suchlike can be used.

[0119] As shown in FIG. 5: [0120] i) a first label group LG1/157A has been determined by the match component **130** and comprises label **165D** (associated with RoI **162D**) from task **1/150A** and label **165G** (associated with RoI **162G**) from task **3/150C**; [0121] ii) a second label group LG2/157B comprises label **165C** (associated with RoI **162C**) from task **1**, label **165E** (associated with RoI **162E**) from task **2/150B**, and label **165I** (associated with RoI **162I**) from task **3**; [0122] iii) a third label group LG3/157C has been determined and comprises label **165A** (associated with RoI **162A**) from task **1**, label **165F** (associated with RoI **162F**) from task **2**, and label **165H** (associated with RoI **162H**) from task **3**; [0123] iv) a fourth label group LG4/157D comprises the single label **165B** (associated with RoI **162B**) from task **1**.

[0124] Hence, while some of the labels **165A-n** across the tasks **150A-n** have a high similarity (e.g., in RoIs **162A-n**) to enable formation of label groups **157A-n**, other labels, such as label **165B** (associated with RoI **162B**), does not have a sufficient degree of matching **510A-n** any other label **165A/165C-n** (associated with RoIs **162A/C-n**) and hence is a sole member of a label group, e.g., LG4/157D.

#### Incremental Learning

[0125] A benefit of utilizing segmentation system **110** and the label sharing model **125A** lies in the ability of the segmentation component **120** and label sharing model **125A** to train the label sharing model **125A** for additional independent anatomies, without necessitating architectural modifications of the label sharing model **125A**. Such an approach can be contingent upon the following conditions: (i) the tasks **150A-n** are independent with no anatomical overlap between them, and (ii) number of labels **165A-n** for the new task **150A** ≤ number of existing channels **260A-n** in the network **125A**.

[0126] Regarding post training with a new task **150A**, catastrophic forgetting for earlier/old/prior/previously applied tasks **150B-n** can be circumvented by conducting combined training of previously applied tasks **150B-n** and new task **150A** through a sequence of iterations, e.g., 5-6 iterations. Such an approach utilizing the label sharing model **125A** process enables satisfactory results to be achieved across all the tasks **150A-n**.

[0127] FIG. 6 presents a sequence of tasks generated by an incrementally trained segmentation model, in accordance with an embodiment. A sequence of tasks **150A-n** are presented indicating results of the incremental learning process. Originally the segmentation model **125A** was trained for landmark/RoI **162A-n** detection in (i) a lower extremity (original task **150A**). The segmentation model **125A** was additionally trained for (ii) landmark/RoI **162A-n** detection in head projection (incremental task **1**, tasks **150B** and **150C**), and further trained for (iii) lung, liver and kidney segmentation on axial slices (incremental task **2**, tasks **150D-F**). Tasks **150A-F** present the inference results from the incrementally trained segmentation model **125A**, with acceptable images generated for tasks **150A-F**, per the various embodiments presented herein. In an embodiment, incremental training can be performed until convergence of at least one of label groups **157A-n** (and included tasks **150A-n**), or the one or more segmented images **196A-n**.

[0128] For the various embodiments presented herein regarding example tasks **150A-n**, the respective imagery, etc., were accomplished using 2D network with **4L** parameters, inferencing at an average time of ~1 s on GPU and ~2.5 s on CPU. The various embodiments can be implemented on any suitable environment, e.g., extended to 2.5D networks, 3D networks, and suchlike.

#### Combined Model Inference Output-after Incremental Learning

[0129] FIG. 7 presents a set of output images indicating no performance degradation resulting from application of the one or more embodiments presented herein. Images **196A-F**, and can be read with the images presented in FIG. 6. Output images **196A-F** were produced as a function of a combined model inference output after the incremental learning process was performed, as previously described, e.g., per FIG. 6. Output images **196A-F** indicate there is no performance degradation on the original task **150A** concerning a lower extremity after incremental learning has been performed on head projection tasks **150B** and **150C** (per FIG. 6, incremental task **1**) and

lung/liver/kidney tasks **150D-F** segmentation (per FIG. **6**, incremental task **2**).

[0130] FIG. **8**, via flowchart **800**, presents an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0131] At **810**, a set of tasks (e.g., tasks **150A-n**) can be received at a segmentation system **110**. The respective tasks can comprise images (e.g., images **160A-n**) having previously applied/annotated labels (e.g., segmentation labels **165A-n**) respectively identifying one or more regions of interest (e.g., RoIs **162A-n**). The labels can be previously annotated by an entity (e.g., a medical expert) having knowledge to identify and label a region of interest. In an embodiment, the sample images **195A-n** utilized to train the segmentation system **110**, segmentation component **120**, match component **130**, process component **170**, processes **176A-n**, and such, can comprise content similar/comparable to the images **160A-n**, RoIs **162A-n**, and labels **165A-n** in tasks **150A-n**, such that sample images **195A-n** can be initially utilized to train any of segmentation component **120**, segmentation model **125A-n**, match component **130**, process component **170**, and/or processes **176A-n**, and once training has been performed, tasks **150A-n** can be subsequently applied to segmentation system **110**.

[0132] At **820**, a match component (e.g., match component **130**) can be configured to identify labels/RoIs on tasks having a degree of similarity such that the respective labels can be grouped (e.g., in label groups **157A-n**). Any suitable measure of similarity can be utilized, e.g., a substantial level of similarity, a threshold level of similarity (e.g., as defined by a threshold measure applied to/implemented at match component **130**), and the like.

[0133] At **830**, the respective tasks and label groupings can be received by a segmentation component (e.g., segmentation component **120**), wherein the segmentation component can apply the tasks and label groupings to a segmentation model (e.g., segmentation model **125A-n**) included in the segmentation component. The segmentation model can be configured with a set of output channels (e.g., output channels **260A-n**), wherein the number of output channels corresponds to the number of label groupings. The segmentation component can be further configured to generate segmented images (e.g., segmented images **196A-n**) comprising respective segmentation masks (e.g., segmentation masks **240A-n**) pertaining to the regions of interest (e.g., RoIs **162A-n**) in the received tasks/images (e.g., images **160A-n** and labels **165A-n**), where, as previously described, the segmentation masks are labeled and associated with a label group (e.g., label group **157A-n**).

[0134] At **840**, application of the tasks and label groupings enable the segmentation model to be trained.

[0135] At **850**, with the segmentation model being modified with the required number of output channels to match the number of label groups, new, unsegmented images (e.g., in subsequently received tasks **150A-n**) can be applied to the trained segmentation model to enable respective regions of interest on the new, unsegmented images to be identified based on the previously applied tasks (e.g., where the previously applied tasks comprise tasks associated with sample images **195A-n** used to train the segmentation component **120**, etc., or received/processed tasks **150A-n** after which the segmentation component **120** undergoes further retraining).

[0136] At **860**, knowledge generated by reviewing the newly segmented images can be reapplied to the training tasks and label groups to facilitate further retraining of the segmentation model as required.

[0137] FIG. **9**, via flowchart **900**, presents an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0138] At **910**, the method **900** can be implemented by a system (e.g., segmentation system **110**), at least one processor, and at least one memory coupled to the at least one processor and having instructions stored thereon, wherein, in response to the at least one processor executing the instructions, the instructions facilitate performance of operations, comprising receiving a set of images (e.g., images **160A-n**), wherein the set of images comprises: a first image (e.g., image **160A**) having a first region of interest (e.g., RoI **162A**), wherein the first image is associated with a

first task (e.g., task **150A**); and a second image (e.g., image **160B**) having a second RoI (e.g., RoI **162B**), wherein the second image is associated with a second task (e.g., task **150B**), and the first RoI and the second RoI are included in a common shared label group (e.g., shared label group **157A**).

[0139] At **920**, the operations of method **900** can further applying the set of images to a segmentation model (e.g., segmentation model **125A**), wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel (e.g., channel **260A**) configured to output segmented images based the common shared label group.

[0140] At **930**, outputting, based on the common shared label group, at least one segmented image (e.g., images **196A-n**) comprising the first RoI or the second RoI.

[0141] FIG. **10**, via flowchart **1000**, presents an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0142] At **1010**, the method **1000** can comprise receiving, by a device (e.g., segmentation system **110**) comprising at least one processor (e.g., processor **182**), a first image (e.g., image **160A**) comprising a first segmentation label (e.g., segmentation **165A**) associated with a first region of interest (e.g., RoI **162A**), wherein the first segmentation label is included in a first shared label group (e.g., shared label group **157A**).

[0143] At **1020**, the method **1000** can further comprise receiving, by the device, a second image, wherein the second image comprises a second RoI (e.g., RoI **162B**) and a second segmentation label (e.g., segmentation label **165B**) associated with the second RoI.

[0144] At **1030**, the method **1000** can further comprise determining, by the device, whether the second RoI relates to the first RoI.

[0145] At **1040**, the method **1000** can further comprise, in response to determining, by the device, the second region of interest is related to the first region of interest, applying, by the device, the second segmentation label to the first common label group.

[0146] FIG. **11**, via flowchart **1100**, presents an example computer-implemented method for utilizing label sharing in an image segmentation process, in accordance with an embodiment.

[0147] At **1110**, the method **1100** can be performed by a computer program product stored on a non-transitory computer-readable medium and comprising machine-executable instructions, wherein, in response to being executed, the machine-executable instructions cause a system to perform operations, comprising receiving a set of images, wherein the set of images comprises: a first image (e.g., image **160A**) having a first region of interest (e.g., RoI **162A**), wherein the first image is associated with a first task, and a second image (e.g., image **160B**) having a second RoI, wherein the second image is associated with a second task (e.g., task **150B**), and the first RoI and the second RoI are included in a common label group (e.g., shared label group **157A**).

[0148] At **1120**, the method **1100** can further comprise applying the set of images to a segmentation model (e.g., segmentation model **125A**), wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel (e.g., channel **260A**) configured to output segmented images (e.g., segmented images **196A-n**) based the common label group.

[0149] At **1130**, the method **1100** can further comprise outputting, based on the common shared label group, at least one segmented image comprising the first RoI or the second RoI.

[0150] As used herein, the terms “infer”, “inference”, “determine”, and suchlike, refer generally to the process of reasoning about or inferring states of the system, environment, and/or user from a set of observations as captured via events and/or data. Inference can be employed to identify a specific context or action, or can generate a probability distribution over states, for example. The inference can be probabilistic—that is, the computation of a probability distribution over states of interest based on a consideration of data and events. Inference can also refer to techniques employed for composing higher-level events from a set of events and/or data. Such inference results in the construction of new events or actions from a set of observed events and/or stored event data, whether or not the events are correlated in close temporal proximity, and whether the events and

data come from one or several event and data sources.

[0151] Per the various embodiments presented herein, various components included in segmentation system **110**, segmentation component **120**, segmentation model **125A-n**, match component **130**, process component **170**, and suchlike, can include AI/ML and reasoning techniques and technologies (e.g., processes **176A-n**) that employ probabilistic and/or statistical-based analysis to prognose or infer an action that a user desires to be automatically performed. The various embodiments presented herein can utilize various machine learning-based schemes for carrying out various aspects thereof. For example, a process **176A-n** (e.g., by match component **130**, segmentation component **120**) for automatically grouping tasks **150A-n** by label groups **157A-n**, automatically modifying segmentation model **125A-n**, and further automatically generating segmented images **196A-n** with the segmentation model **125A-n**, and suchlike, as previously mentioned herein, can be facilitated via an automatic classifier system and process.

[0152] A classifier is a function that maps an input attribute vector,  $x=(x_1, x_2, x_3, x_4, x_n)$ , to a class label  $\text{class}(x)$ . The classifier can also output a confidence that the input belongs to a class, that is,  $f(x)=\text{confidence}(\text{class}(x))$ . Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to prognose or infer an action that a user desires to be automatically performed (e.g., identifying label groups **157A-n**, and operations related thereto).

[0153] A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs that splits the triggering input events from the non-triggering events in an optimal way. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches include, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein is inclusive of statistical regression that is utilized to develop models of priority.

[0154] As will be readily appreciated from the subject specification, the various embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing user behavior, receiving extrinsic information). For example, SVM's are configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria, content in tasks **150A-n**, images **160A-n**, RoIs **162A-n**, labels **165A-n**, label groups **157A-n**, sample images **195A-n**, and modify the segmentation model **125A-n** (e.g., with correct number of output channels **260A-n** for example).

[0155] As described supra, inferences can be made, and automated operations performed, based on numerous pieces of information. For example, whether sufficient context is available to infer, with a high degree of confidence, a correlation between labels **165A-n**, and associated RoIs **162A-n**, to generate the label groups **157A-n**, modify the segmentation model **125A-n**, and suchlike.

#### Example Applications and Use

[0156] Turning next to FIGS. **12** and **13**, a detailed description is provided of additional context for the one or more embodiments described herein with FIGS. **1-11**.

[0157] In order to provide additional context for various embodiments described herein, FIG. **12** and the following discussion are intended to provide a brief, general description of a suitable computing environment **1200** in which the various embodiments described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0158] Generally, program modules include routines, programs, components, data structures, etc.,



that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, IoT devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0159] The embodiments illustrated herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0160] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

[0161] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0162] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0163] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0164] With reference again to FIG. 12, the example environment **1200** for implementing various embodiments of the aspects described herein includes a computer **1202**, the computer **1202** including a processing unit **1204**, a system memory **1206** and a system bus **1208**. The system bus **1208** couples system components including, but not limited to, the system memory **1206** to the processing unit **1204**. The processing unit **1204** can be any of various commercially available processors and may include a cache memory. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **1204**.

[0165] The system bus **1208** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local

bus using any of a variety of commercially available bus architectures. The system memory **1206** includes ROM **1210** and RAM **1212**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1202**, such as during startup. The RAM **1212** can also include a high-speed RAM such as static RAM for caching data.

[0166] The computer **1202** further includes an internal hard disk drive (HDD) **1214** (e.g., EIDE, SATA), one or more external storage devices **1216** (e.g., a magnetic floppy disk drive (FDD) **1216**, a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive **1220** (e.g., which can read or write from a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD **1214** is illustrated as located within the computer **1202**, the internal HDD **1214** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1200**, a solid-state drive (SSD) could be used in addition to, or in place of, an HDD **1214**. The HDD **1214**, external storage device(s) **1216** and optical disk drive **1222** can be connected to the system bus **1208** by an HDD interface **1224**, an external storage interface **1226** and an optical drive interface **1228**, respectively. The interface **1224** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0167] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1202**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0168] A number of program modules can be stored in the drives and RAM **1212**, including an operating system **1230**, one or more application programs **1232**, other program modules **1234** and program data **1236**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1212**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0169] Computer **1202** can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system **1230**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. **12**. In such an embodiment, operating system **1230** can comprise one virtual machine (VM) of multiple VMs hosted at computer **1202**. Furthermore, operating system **1230** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1232**. Runtime environments are consistent execution environments that allow applications **1232** to run on any operating system that includes the runtime environment. Similarly, operating system **1230** can support containers, and applications **1232** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0170] Further, computer **1202** can comprise a security module, such as a trusted processing module (TPM). For instance with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1202**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0171] A user can enter commands and information into the computer **1202** through one or more

wired/wireless input devices, e.g., a keyboard **1238**, a touch screen **1240**, and a pointing device, such as a mouse **1242**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1204** through an input device interface **1244** that can be coupled to the system bus **1208**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

[0172] A monitor **1246** or other type of display device can be also connected to the system bus **1208** via an interface, such as a video adapter **1248**. In addition to the monitor **1246**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0173] The computer **1202** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1250**. The remote computer(s) **1250** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1202**, although, for purposes of brevity, only a memory/storage device **1252** is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) **1254** and/or larger networks, e.g., a wide area network (WAN) **1256**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the internet.

[0174] When used in a LAN networking environment, the computer **1202** can be connected to the local network **1254** through a wired and/or wireless communication network interface or adapter **1258**. The adapter **1258** can facilitate wired or wireless communication to the LAN **1254**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1258** in a wireless mode.

[0175] When used in a WAN networking environment, the computer **1202** can include a modem **1260** or can be connected to a communications server on the WAN **1256** via other means for establishing communications over the WAN **1256**, such as by way of the internet. The modem **1260**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1208** via the input device interface **1244**. In a networked environment, program modules depicted relative to the computer **1202** or portions thereof, can be stored in the remote memory/storage device **1252**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0176] When used in either a LAN or WAN networking environment, the computer **1202** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **1216** as described above. Generally, a connection between the computer **1202** and a cloud storage system can be established over a LAN **1254** or WAN **1256** e.g., by the adapter **1258** or modem **1260**, respectively. Upon connecting the computer **1202** to an associated cloud storage system, the external storage interface **1226** can, with the aid of the adapter **1258** and/or modem **1260**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **1226** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **1202**.

[0177] The computer **1202** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location

associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0178] The above description includes non-limiting examples of the various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the disclosed subject matter, and one skilled in the art may recognize that further combinations and permutations of the various embodiments are possible. The disclosed subject matter is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims.

[0179] Referring now to details of one or more elements illustrated at FIG. 13, an illustrative cloud computing environment **1300** is depicted. FIG. 13 is a schematic block diagram of a computing environment **1300** with which the disclosed subject matter can interact. The system **1300** comprises one or more remote component(s) **1310**. The remote component(s) **1310** can be hardware and/or software (e.g., threads, processes, computing devices). In some embodiments, remote component(s) **1310** can be a distributed computer system, connected to a local automatic scaling component and/or programs that use the resources of a distributed computer system, via communication framework **1340**. Communication framework **1340** can comprise wired network devices, wireless network devices, mobile devices, wearable devices, radio access network devices, gateway devices, femtocell devices, servers, etc.

[0180] The system **1300** also comprises one or more local component(s) **1320**. The local component(s) **1320** can be hardware and/or software (e.g., threads, processes, computing devices). In some embodiments, local component(s) **1320** can comprise an automatic scaling component and/or programs that communicate/use the remote resources **1310** and **1320**, etc., connected to a remotely located distributed computing system via communication framework **1340**.

[0181] One possible communication between a remote component(s) **1310** and a local component(s) **1320** can be in the form of a data packet adapted to be transmitted between two or more computer processes. Another possible communication between a remote component(s) **1310** and a local component(s) **1320** can be in the form of circuit-switched data adapted to be transmitted between two or more computer processes in radio time slots. The system **1300** comprises a communication framework **1340** that can be employed to facilitate communications between the remote component(s) **1310** and the local component(s) **1320**, and can comprise an air interface, e.g., Uu interface of a UMTS network, via a long-term evolution (LTE) network, etc. Remote component(s) **1310** can be operably connected to one or more remote data store(s) **1350**, such as a hard drive, solid state drive, SIM card, device memory, etc., that can be employed to store information on the remote component(s) **1310** side of communication framework **1340**. Similarly, local component(s) **1320** can be operably connected to one or more local data store(s) **1330**, that can be employed to store information on the local component(s) **1320** side of communication framework **1340**.

[0182] With regard to the various functions performed by the above described components, devices, circuits, systems, etc., the terms (including a reference to a “means”) used to describe such components are intended to also include, unless otherwise indicated, any structure(s) which performs the specified function of the described component (e.g., a functional equivalent), even if not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosed subject matter may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

[0183] The terms “exemplary” and/or “demonstrative” as used herein are intended to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as

“exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent structures and techniques known to one skilled in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive—in a manner similar to the term “comprising” as an open transition word—without precluding any additional or other elements.

[0184] The term “or” as used herein is intended to mean an inclusive “or” rather than an exclusive “or.” For example, the phrase “A or B” is intended to include instances of A, B, and both A and B. Additionally, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless either otherwise specified or clear from the context to be directed to a singular form.

[0185] The term “set” as employed herein excludes the empty set, i.e., the set with no elements therein. Thus, a “set” in the subject disclosure includes one or more elements or entities. Likewise, the term “group” as utilized herein refers to a collection of one or more entities.

[0186] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0187] As used in this disclosure, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component.

[0188] One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components can communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software application or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0189] The term “facilitate” as used herein is in the context of a system, device or component “facilitating” one or more actions or operations, in respect of the nature of complex computing environments in which multiple components and/or multiple devices can be involved in some computing operations. Non-limiting examples of actions that may or may not involve multiple components and/or multiple devices comprise transmitting or receiving data, establishing a connection between devices, determining intermediate results toward obtaining a result, etc. In this

regard, a computing device or component can facilitate an operation by playing any part in accomplishing the operation. When operations of a component are described herein, it is thus to be understood that where the operations are described as facilitated by the component, the operations can be optionally completed with the cooperation of one or more other computing devices or components, such as, but not limited to, sensors, antennae, audio and/or visual output devices, other devices, etc.

[0190] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable (or machine-readable) device or computer-readable (or machine-readable) storage/communications media. For example, computer readable storage media can comprise, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0191] Moreover, terms such as “mobile device equipment,” “mobile station,” “mobile,” “subscriber station,” “access terminal,” “terminal,” “handset,” “communication device,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or mobile device of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings. Likewise, the terms “access point (AP),” “Base Station (BS),” “BS transceiver,” “BS device,” “cell site,” “cell site device,” “gNode B (gNB),” “evolved Node B (eNode B, eNB),” “home Node B (HNB)” and the like, refer to wireless network components or appliances that transmit and/or receive data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream from one or more subscriber stations. Data and signaling streams can be packetized or frame-based flows.

[0192] Furthermore, the terms “device,” “communication device,” “mobile device,” “subscriber,” “client entity,” “consumer,” “client entity,” “entity” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0193] It should be noted that although various aspects and embodiments are described herein in the context of 5G or other next generation networks, the disclosed aspects are not limited to a 5G implementation, and can be applied in other network next generation implementations, such as sixth generation (6G), or other wireless systems. In this regard, aspects or features of the disclosed embodiments can be exploited in substantially any wireless communication technology. Such wireless communication technologies can include universal mobile telecommunications system (UMTS), global system for mobile communication (GSM), code division multiple access (CDMA), wideband CDMA (WCDMA), CDMA2000, time division multiple access (TDMA), frequency division multiple access (FDMA), multi-carrier CDMA (MC-CDMA), single-carrier CDMA (SC-CDMA), single-carrier FDMA (SC-FDMA), orthogonal frequency division multiplexing (OFDM), discrete Fourier transform spread OFDM (DFT-spread OFDM), filter bank based multi-carrier (FBMC), zero tail DFT-spread-OFDM (ZT DFT-s-OFDM), generalized frequency division multiplexing (GFDM), fixed mobile convergence (FMC), universal fixed mobile convergence (UFMC), unique word OFDM (UW-OFDM), unique word DFT-spread OFDM (UW DFT-Spread-OFDM), cyclic prefix OFDM (CP-OFDM), resource-block-filtered OFDM, wireless fidelity (Wi-Fi), worldwide interoperability for microwave access (WiMAX), wireless local area network

(WLAN), general packet radio service (GPRS), enhanced GPRS, third generation partnership project (3GPP), long term evolution (LTE), 5G, third generation partnership project 2 (3GPP2), ultra-mobile broadband (UMB), high speed packet access (HSPA), evolved high speed packet access (HSPA+), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Zigbee, or another institute of electrical and electronics engineers (IEEE) 802.12 technology.

[0194] It is to be understood that when an element is referred to as being “coupled” to another element, it can describe one or more different types of coupling including, but not limited to, chemical coupling, communicative coupling, electrical coupling, electromagnetic coupling, operative coupling, optical coupling, physical coupling, thermal coupling, and/or another type of coupling. Likewise, it is to be understood that when an element is referred to as being “connected” to another element, it can describe one or more different types of connecting including, but not limited to, electrical connecting, electromagnetic connecting, operative connecting, optical connecting, physical connecting, thermal connecting, and/or another type of connecting.

[0195] The description of illustrated embodiments of the subject disclosure as provided herein, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as one skilled in the art can recognize. In this regard, while the subject matter has been described herein in connection with various embodiments and corresponding drawings, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

## Claims

1. A system, comprising: at least one processor; and at least one memory coupled to the at least one processor and having instructions stored thereon, wherein, in response to the at least one processor executing the instructions, the instructions facilitate performance of operations, comprising: receiving a set of images, wherein the set of images comprises: a first image having a first region of interest (RoI), wherein the first image is associated with a first task; and a second image having a second RoI, wherein the second image is associated with a second task, and the first RoI and the second RoI are included in a common shared label group; applying the set of images to a segmentation model, wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel configured to output segmented images based the common shared label group; and outputting, based on the common shared label group, at least one segmented image comprising the first RoI or the second RoI.
2. The system of claim 1, wherein the common label group is a first label group, the output channel is a first output channel, and at least one segmented image is included in a first set of segmented images, wherein the operations further comprise: receiving a second set of images sharing a second shared label group, wherein the first shared label group and the second shared label group are disparate, and the second set of images comprises: a third image having a third RoI, wherein the third image is associated with the first task; and a fourth image having a fourth RoI, wherein the fourth image is associated with the second task, and the third RoI and the fourth RoI are included in a second shared label group; applying the second set of images to the segmentation model, wherein the segmentation model further comprises a second output channel configured to output segmented images based on the second shared label group; and outputting, based on the second shared label

group, at least one segmented image comprising the third RoI or the fourth RoI.

3. The system of claim 2, wherein an architecture of the segmentation model is common to application of two or more segmentation tasks and generation of segmented images based on the first shared label group and the second shared label group.
4. The system of claim 2, wherein a set of common weightings are applied during creation of both the first shared label group and the second shared label group during application of two or more segmentation tasks.
5. The system of claim 2, wherein labels in the first shared label group are disparate to the labels in the second shared label group.
6. The system of claim 2, wherein the operations further comprise: determining a first number of shared label groups in the first task and a second number of shared label groups in the second task; and in response to a determination that the first number of shared label groups is different to the second number of shared label groups, configuring the number of output channels of the segmentation model to equal the greater of the first number of shared label groups or the second number of shared label groups.
7. The system of claim 1, wherein the operations further comprise generating the common shared label group as a function of at least one similarity between the at least two images.
8. The system of claim 1, wherein the operations further comprise: applying a third task to the segmentation model; generating at least one shared label group from the third task, wherein the at least one shared label group includes the common shared label group; and training the segmentation model across multiple tasks incrementally utilizing any of the at least two images or the common shared label group.
9. The system of claim 8, wherein the operations further comprise: subsequently applying a subsequent task to the segmentation model to facilitate incremental training of the segmentation model over a series of iterations to adapt the segmentation model to the subsequent task; and maintaining incremental training until determining convergence has occurred.
10. The system of claim 9, wherein the operations further comprise: re-applying a prior task to the converged segmentation model to prevent catastrophic forgetting of the prior task as a function of the segmentation model being converged with the subsequent task.
11. A computer-implemented method, comprising: receiving, by a device comprising at least one processor, a first image comprising a first segmentation label associated with a first region of interest (RoI), wherein the first segmentation label is included in a first shared label group; receiving, by the device, a second image, wherein the second image comprises a second RoI and a second segmentation label associated with the second RoI; determining, by the device, whether the second RoI relates to the first RoI; and in response to determining, by the device, the second RoI is related to the first RoI, applying, by the device, the second segmentation label to the first shared label group.
12. The computer-implemented method of claim 11, wherein the first shared label group is associated with a first channel assigned to a segmentation model implemented by the device, wherein the segmentation model is implemented by the device to determine the second RoI is related to the first RoI.
13. The computer-implemented method of claim 12, wherein, in response to determining by the device, the second RoI is unrelated to the first RoI, the computer-implemented method further comprising: identifying, by the device, a third image comprising a third segmentation label, wherein the third segmentation label is associated with a third RoI and the third segmentation label is included in a second shared label group; determining, by the device, the second RoI relates to the third RoI; and applying the second segmentation label to the second shared label group, wherein the second shared label group is associated with a second channel assigned to the segmentation model implemented by the device, wherein the segmentation model is implemented by the device to determine the second RoI is related to the third RoI.



- 14.** The computer-implemented method of claim 13, wherein, in response to determining by the device, the second RoI is unrelated to the first RoI and is also unrelated to the third RoI, the computer-implemented method further comprising: updating, by the device, the segmentation model to include a third channel, wherein the third channel is assigned with a third shared label group; and assigning the second segmentation label to the third shared label group.
- 15.** The computer-implemented method of claim 12, wherein the segmentation model is a U-net model.
- 16.** The computer-implemented method of claim 11, wherein the determination, by the device, of the second RoI relates to the first RoI is based on at least one of similarity of shape of the first RoI to the second RoI, pixel level between the first RoI and the second RoI, or patch level between a first region of interest (RoI) and a secondRoI.
- 17.** A computer program product stored on a non-transitory computer-readable medium and comprising machine-executable instructions, wherein, in response to being executed, the machine-executable instructions cause a system to perform operations, comprising: receiving a set of images, wherein the set of images comprises: a first image having a first region of interest (RoI), wherein the first image is associated with a first task; and a second image having a second RoI, wherein the second image is associated with a second task, and the first RoI and the second RoI are included in a common label group; applying the set of images to a segmentation model, wherein the segmentation model is a computer-implemented segmentation model and comprises an output channel configured to output segmented images based the common label group; and outputting, based on the common shared label group, at least one segmented image comprising the first RoI or the second RoI.
- 18.** The computer program product of claim 17, wherein the set of images in the task comprise medical images.
- 19.** The computer program product of claim 17, wherein the first RoI being related to the second RoI is based on at least one of similarity of shape of the first RoI to the second RoI, pixel level between the first RoI and the second RoI, or patch level between a first RoI and a second RoI.
- 20.** The computer program product of claim 17, wherein the operations further comprising defining the number of output channels prior to generating the set of shared label groups.
-